Resilience Enhancement for a Power System with Electric Vehicles under Extreme Weather Conditions

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***Abstract*—In this paper, a resilience enhancement model is presented for a power system with plug-in electric vehicles (EVs) under extreme weather conditions. The proposed methodological framework is formulated as a two stage/tri-level robust mixed-integer optimization model. In order to solve this model, an algorithm based on nested column-and-constraint generation (NC&CG) is developed. Numerical simulations of a sample power system validate the effectiveness of the developed framework and the efficiency of the solving methodology.**

***Index Terms*—power system, resilience enhancement, electric vehicle, load shedding, robust optimization, normal stage, resilience stage, nested column-and-constraint generation (NC&CG).**

I. NOMENCLATURE

A. Index

ei	Index of EVs
gi	Index of generator
Di	Index of load
di	Index of shedding load
i, j	Index of buses

B. Parameters, Variables

C_{gi}	The offer price of generator g
k	Maximum number of damaged lines
$E^{t}_{EVS,avail}$	Total energy storage level for the available EVs in time t
$E_{EVS,total}$	Total rated energy storage level for all EVs
P_{di}	Load shedding power
$N_{EV.avail}$	The number of EVs in charging/swapping stations
P_{EV}	Active power consumed in EVs charging/swapping stations
$P^{t}_{EV.CH}$	Maximum charging power in EVs charging/swapping stations at time *t*
$P^{t}_{EV.DIS}$	Maximum discharging power in EVs charging/swapping stations at time *t*
$\overline{P}_{ev.ch(ei)}$	Maximum charging power of EVs
$\overline{P}_{ev.dis(ei)}$	Maximum discharging power of EVs
P_{Di}	Power demand of load
$P_{ij,t}$	Active power on a transmission line
$P^{\max}_{ij}$	Transmission line capacity
P_{gi}	Conventional generator power output
$P^{\max}_{gi}$	Maximal generator power output
η_{ch}	The charging parameter of EVs charging/swapping station
η_{dis}	The discharging parameter of EVs charging/swapping station
θ_i	Bus angle
k	Maximum number of power line outages due to extreme events
x_{ij}	Reactance of power lines
u_{ij}	State variable representing existing power lines
z_{ij}	State variable representing power line outage

II. INTRODUCTION

Since the 21st century, many countries have accelerated the development of the power industry at different levels, such as regional or even national networking, new energy development, grid-connected power generation, and the wide applications of a large number of electric vehicles.

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Therefore, the installation components, structure and coordination control of the power system are more complex, and security, stability and economic issues are more prominent. At the same time, extreme weather and natural disasters, as well as man-made damages and terrorist attacks have further increased the uncertainty about the development of the power system, which is still subject to massive blackout threats [1-3]. The electric power system is responsible for the electric energy generation and the transmission infrastructure. Large-scale power failure, not only will cause huge economic losses, but also may cause political, social impacts and even human casualties.

Under this background, many electric power companies, research institutions and universities have carried out such research work on the development mode of the future electric power system and proposed the concept of the resilient power system[1].

The so-called resilient power system refers to the power system with both adaptive and flexible control and strong impacts resistance. The system can ensure safe and stable operation in case of large disturbance under normal operation or short-circuit conditions. When the system encounters disturbance events, it has the ability to make corresponding preparation and prevention. In the context of a power system, resilience represents the ability to withstand low-probability but high-impact incidents in an efficient manner while ensuring the least possible interruption in supply of electricity, and further enabling a quick restoration to the normal operation state [4].

In existing publications, many researchers have shown great interests to improve the power grid resistance against extreme events. A resilience-oriented design (ROD) technique to protect distribution grids against high-impact but low-probability extreme weather events was developed by Shanshan Ma et al. in [5]. N. Alguacil et al. proposed a new planning model for the optimal allocation of defensive resources in a power grid to mitigate the vulnerability against multiple contingencies [6]. An integrated resilience response framework was studied to integrate the preventive stage and emergency stage of the resilience response [7]. A proactive operation strategy to enhance system resilience during an unfolding extreme event was developed. The uncertain sequential transition of system states driven by the evolution of extreme events is modeled as a Markov process [8]. After an extreme weather event related power outage, resilience restoration plays a vital role in doing away with the system restoration and load restoration and it's presumed that resilience restoration could largely reduce weather disaster related economic loss [9,10].

In order to increase the resilience of a power system, one popular approach is to use a defender-attacker-defender model, which also takes form of two-stage robust optimization formulation [11-12]. In the multi-level optimization structure, the power system operator decides a distribution expansion plan in the first stage. In the second stage of the model, an extreme natural disaster disrupts on the power grid aiming at the maximum damage. Finally, as the defender, the power system operator, reacts to the extreme event by adjusting system power flow and load shedding [13].

However, other most immediate and essential factors that has given rise to power system resilience enhancement, in particular when people have to take into consideration the extreme events, were ignored in those research works. In the past 10 years, some profound changes have taken place in the power industry. The smart grid construction featuring high permeability intermittent renewable energy generation and wide application of electric vehicles has become the main development direction of modern power system [14]. People can make use of electric vehicles based on power electronics, modern advanced communication and control technology to enhance the ability to resist major natural disasters, improve the system self-adaptation and fault self-healing ability, and ensure the safe and stable operation of the power system.

As flexible schedulable demand side resources, EVs can be used in power system operation, such as peak load shifting, providing ancillary services, suppressing the energy fluctuations of renewable energy, and so on. EVs battery is a kind of flexible energy storage unit in power system, and it can participate in the resilience enhancement of power system after the power failure accidents. There are two major modes of EVs charging, namely concentrated charging and batteries swapping [15]. Compared with the home-charging mode, a large number of EVs batteries could be dispatched in the mode of concentrated charging and batteries swapping. As a result, EVs can play a key role in resilience enhancement of power grid.

The rest of this paper is organized as follows. The characteristics of EVs in resilience are proposed and the uncertain set of damaged transmission lines is introduced to describe the influence of worst weather events in Section II. An integrated resilience enhancement framework for a power system with EVs is developed in Section III. A modified NC&CG algorithm is provided in Section IV. Afterwards, comprehensive case studies are performed in Section V. Section VI finally concludes the paper.

III. RESILIENCE ENHANCEMENT STRATEGIES

A. Characteristics of Electric Vehicles in Resilience Enhancement

In the initial stage of power system restoration, it is generally necessary to restore some no-load or light-load transmission lines. EVs can relieve the overvoltage generated by the line charging. In the meantime, the starting power should provide for the non-black starting units by the black startup units. Then, the non-black startup units need to be stable enough in the minimum output power level. Feeding controllable EVs into a power system will help the power system to maintain the active power real-time balance and accelerate the restoration process of the power system concerned.

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Generally speaking, it is convenient for the power system dispatch center to schedule EVs charging and discharging by using the modes of concentrated charging and batteries swapping. On the contrary, the home charging mode is much more decentralized and difficult to control. In this paper, the number of storage batteries in swapping station is regarded as the number of EVs, which can be considered with union of the concentrated charging station.

The charging/discharging maximum power of EVs concentrated charging/swapping station is in follows:

$$P^{t}_{EV.CH} = \sum_{ei=1}^{N^{t}_{EV.avail}} \overline{P}_{ev.ch(ei)} \tag{1}$$

$$P^{t}_{EV.DIS} = \sum_{ei=1}^{N^{t}_{EV.avail}} \overline{P}_{ev.dis(ei)} \tag{2}$$

However, if the EVs battery energy storage approximates to its upper or lower limits, the charging/discharging maximum power of EVs battery will be limited to battery capacity. To deal with this issue, we introduce charging and discharging power adjustment factors, η_{ch} and η_{dis} as follows:

$$\eta^{t}_{ch} = \left(1 - \frac{E^{t}_{EVS,avail}}{E_{EVS.total}}\right)^{\frac{1}{2}} \tag{3}$$

$$\eta^{t}_{dis} = \left(\frac{E^{t}_{EVS,avail}}{E_{EVS.total}}\right)^{\frac{1}{2}} \tag{4}$$

Then (1)-(2) become:

$$P^{t}_{EV.CH} = \eta^{t}_{ch} \sum_{ei=1}^{N^{t}_{EV.avail}} \overline{P}_{ev.ch(ei)} \tag{5}$$

$$P^{t}_{EV.DIS} = \eta^{t}_{dis} \sum_{ei=1}^{N^{t}_{EV.avail}} \overline{P}_{ev.dis(ei)} \tag{6}$$

Where $E^{t}_{EVS,avail}$ represents the total energy storage level for the available EVs in time t, and $E_{EVS,total}$ represents the total rated energy storage level for all EVs. When $E^{t}_{EVS,avail}$ is high, there will be more EVs batteries to reach saturation or close to saturation, then the number of rechargeable EVs will be lower, and the number of dischargeable EVs will be higher, and vice versa.

B. Damage from Disasters

In this paper, the impact of worst weather scenarios is measured by the uncertain set of damaged transmission lines because they are the most likely damaged electricity components [16]. The uncertain set of damaged lines can be modeled as,

$$Z = \left\{ z_{ij} \mid \sum_{ij} z_{ij} \le k \right\} \tag{7}$$

Where k is maximum number of power line outages due to extreme events. There is no doubt that damaged transmission lines must occur at existing power lines. So

$$z_{ij} \le u_{ij} \tag{8}$$

IV. Mathematical Formulation

In this paper, we developed a two-stage/tri-level resilience enhancement model as shown in Fig. 1. The objective is to minimize the power system operation cost and load shedding cost under the worst-case scenarios. The first level is to make the power system operate in an economic point of view. In order to enhance the power system resilience, the load shedding cost in the resilience stage should also be considered in the normal operation stage. In the second level, extreme events are considered and the damage of the transmission lines is maximized. After the natural disaster unfold, power system operators react to the extreme events by adjusting the system loads in the third level.

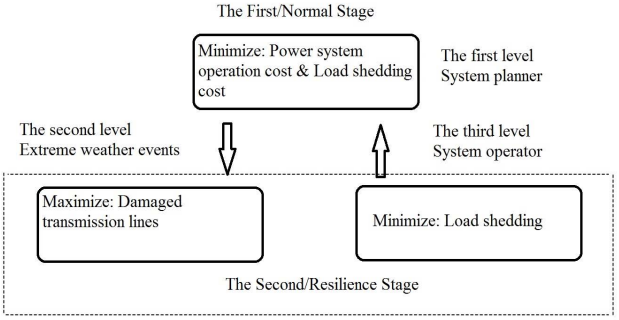


Fig.1. The developed two-stage/tri-level model for power system resilience enhancement

A. Normal Operation Stage

1).Objective Function

The objective of the proposed formulation is to minimize power system operation cost in normal operation stage and the projected load shedding cost under the worst weather events, stated as

$$Min \sum_{gi} C_{gi} P^{a}_{gi} + \sum_{di} C_{di} P^{b}_{di} \tag{9}$$

2).System Constraints

In the initial stage, the extreme weather scenarios are not considered. As a result, the power system will supply the initial demand without considering any load curtailments. The operation constraints of normal stage are presented as follows.

Power system constraints

$$\sum_{gi} P^{a}_{gi} - \sum_{Li} P^{a}_{ij} - \sum_{Di} P_{Di} - P^{a}_{EV} = 0 \tag{10}$$

$$p^{a}_{ij,t} - \frac{\theta^{a}_{i} - \theta^{a}_{j}}{x_{ij}} + M\,(1 - u^{a}_{ij}) \ge 0 \tag{11}$$

$$p^{a}_{ij,t} - \frac{\theta^{a}_{i} - \theta^{a}_{j}}{x_{ij}} - M\,(1 - u^{a}_{ij}) \le 0 \tag{12}$$

$$-p^{\max}_{ij} u^{a}_{ij} \le p^{a}_{ij} \le p^{\max}_{ij} u^{a}_{ij} \tag{13}$$

$$\theta^{\min}_{i} \le \theta^{a}_{i} \le \theta^{\max}_{i} \tag{14}$$

Constraint (10) represents the power balance at each node, and constraints (11) & (12) represent the power balance for each transmission line. We apply the DC power flow equations. Constraint (13) enforces the transmission line flow limits and each bus voltage angle should be within the specified limits as presented in constraint (14).

Generation Unit Constraints

$$0 \le p^{a}_{gi,t} \le p^{\max}_{gi} \tag{15}$$

Constraint (15) limits active power output of generators.

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EVs constraints

$$P^{a}_{EV.CH} = \eta^{a}_{ch} \sum_{ei}^{N^{a}_{EV_avail}} \overline{P}_{ev.ch(ei)} \tag{16}$$

$$P^{a}_{EV.DIS} = \eta^{a}_{dis} \sum_{ei}^{N^{a}_{EV_avail}} \overline{P}_{ev.dis(ei)} \tag{17}$$

$$-p^{a}_{EV.DIS} \le P^{a}_{EV} \le P^{a}_{EV.CH} \tag{18}$$

Electrical vehicles (EVs) constraints are given in (16)-(18). The charging/discharging maximum power of EVs concentrated charging/swapping station in normal stage is respectively given in (16) and (17). Constraint (18) imposes the active power consumed limits in EVs charging/swapping stations.

B. Resilience Stage

1).Objective Function

At the second/resilience stage, worst weather scenarios are considered and the Resilience Metric (RM) is developed by

$$RM = \underset{z_{ij}}{Max}\ \underset{P}{Min} \sum_{di} P^{b}_{di} \le RM_{\max} \tag{19}$$

The difference between total load demand and the critical load demand is presented as $RM_{\max}$. In practice, it can be roughly thought of as the amount of critical loads, such as loads for governments and security facilities.

In the inner minimization, power system operators minimize the load shedding due to the worst disaster events. In the outside maximization, the damaged transmission lines during the extreme weather are maximized.

2).System Constraints

The corresponding second stage constraints are improved in (20)-(24), which are similar to the system constraints in the first stage.

Power system constraints

$$\sum_{gi} P^{b}_{gi} - \sum_{Li} P^{b}_{ij} - \sum_{Di} P_{Di} + \sum_{di} P^{b}_{di} - P^{b}_{EV} = 0 \tag{20}$$

$$p^{b}_{ij} - \frac{\theta^{b}_{i} - \theta^{b}_{j}}{x_{ij}} + M\,(1 - u^{b}_{ij} + z_{ij}) \ge 0 \tag{21}$$

$$p^{b}_{ij,t} - \frac{\theta^{b}_{i} - \theta^{b}_{j}}{x_{ij}} - M\,(1 - u^{b}_{ij} + z_{ij}) \le 0 \tag{22}$$

$$-p^{\max}_{ij}(u^{b}_{ij} - z_{ij}) \le p^{b}_{ij} \le p^{\max}_{ij}(u^{b}_{ij} - z_{ij}) \tag{23}$$

$$\theta^{\min}_{i} \le \theta^{b}_{i} \le \theta^{\max}_{i} \tag{24}$$

The power balance of each node is modified in constraint (20) and the power balance of transmission lines is developed in (21) and (22). Transmission line flow limits constraint in the resilience stage is revised in (23), and bus voltage angle limits constraint in (24) is similar to that in the first stage.

Generation Unit Constraints

$$0 \le p^{b}_{gi} \le p^{\max}_{gi} \tag{25}$$

Similar to the normal stage, each generator capacity is limited in constraint (25).

EVs Constraints

$$P^{b}_{EV.CH} = \eta^{b}_{ch} \sum_{ei=1}^{N^{b}_{EV_avail}} \overline{P}_{ev.ch(ei)} \tag{26}$$

$$P^{b}_{EV.DIS} = \eta^{b}_{dis} \sum_{ei=1}^{N^{b}_{EV_avail}} \overline{P}_{ev.dis(ei)} \tag{27}$$

$$-p^{b}_{EV.DIS} \le P^{b}_{EV} \le P^{b}_{EV.CH} \tag{28}$$

EVs constraints are similar to the normal stage.

C. Final Formulation

Based on the previous statement, the two stage/tri-level resilience enhancement model can be given as follows:

$$\begin{aligned} &\min\ (9) \\ &s.t.\ \ (10)-(18) \\ &\quad \max \min (19) \\ &\quad\quad s.t.(20)-(28) \end{aligned} \tag{29}$$

In addition, the resilience model can be further expressed as,

$$\begin{aligned} &\min_{p^{0}, p_{di}} F^{0}(p^{0}, p_{di}) \\ &s.t.\ \ A^{0}p^{0} \le B^{0}u^{0} \\ &\quad \max_{z}\ \min_{p}\ (p) \le RM_{\max} \\ &\quad\quad Ap \le Bu + cz \\ &\quad\quad z \in Z(u) \end{aligned} \tag{30}$$

The proposed model is a tri-level MILP problem with linear generation cost and load shedding cost. In the next section, we will develop a solution algorithm to solve it efficiently.

V. Solution Algorithm

A. Problem Reformulation

The proposed resilience model is a tri-level mixed-integer linear optimization problem, which can be solved by nested column-and-constraint generation (NC&CG) decomposition framework [17]. A solution based on the NC&CG decomposition can detect the uncertainty outages and add the corresponding constraints into the original problem to replace the uncertainty set. Then the original model is decomposed into a master problem and a sub-problem in (31) and (32) respectively. The uncertainty outages set is replaced by a series of detected worst events $\{z^{\wedge s}\}$, and the master problem is a relaxation of the original one. Based on the solution of the master problem $p^{\wedge s}_{di}$, the sub-problem is proposed to calculate the resilience metric. The solution of sub-problem $z^{\wedge s}$ will be added to the worst weather scenarios $\{z^{\wedge s}\}$. The algorithm iteratively generates new solutions of the master problem and sub-problem until the resilience metric is smaller than $RM_{\max}$.

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$$\min_{p^0, p_{di}} F^0(p^0, p_{di})$$
$$s.t.\ \ A^0 p^0 \le B^0 u^0 \qquad (31)$$
$$p^s \le RM_{\max}$$
$$Ap^s \le Bu^0 + Cz^{\wedge s} (s = 1,2,3...)$$

$$\max_{z} \min_{p} (p)$$
$$s.t. \quad Ap \le Bu^{\wedge s} + Cz \qquad (32)$$
$$z \in Z(u^{\wedge s})$$

Then, the sub-problem is converted into a bilinear problem by the duality-based algorithm of the inner minimization. Furthermore, the sub-problem is reformulated as a MILP problem by the Big-M method. The master problem is also a MILP problem, so both of them can be solved by CPLEX or GUROBI.

B. Solution Algorithm

The detailed procedures for the developed algorithm are described in the below framework.

Solution Algorithm

Step 0: Initialization. Set the resilience metric $RM = 0$, iteration count s=0, worst weather event scenarios $z^{\wedge s} = 0$. Moreover, we know the initial generators outputs p^0 and the initial number of existing power lines u^0.

Step 1: While $RM > RM_{\max}$, **do**

(1). Solve the master problem and calculate the optimal load shedding $p_{di}^{\wedge s}$;

(2). Solve the sub-problem and determine the optimal solution $z^{\wedge s}$ and update the resilience metric RM;

(3). Based on $z^{\wedge s}$ form the enabling constraints stated in (7) and (8), create generators dispatch variable p^s and the corresponding operation constraints, add the variables and constraints to the master problem;

(4). Update $s = s + 1$;

Step 2: end do

Step 3: return $p_{di}^* = p_{di}^{\wedge s}$

VI. CASE STUDIES

Modified IEEE One-Area RTS-96 System

The proposed solution algorithm is implemented on a modified version of the IEEE RTS-96 system [18]. This system consists of 12 generators, 2 EVs charging/swapping stations in node 6 and 19 respectively, 34 transmission lines and 20 system loads. In this case, $RM_{\max}$ is chosen as 5% of the electricity load.

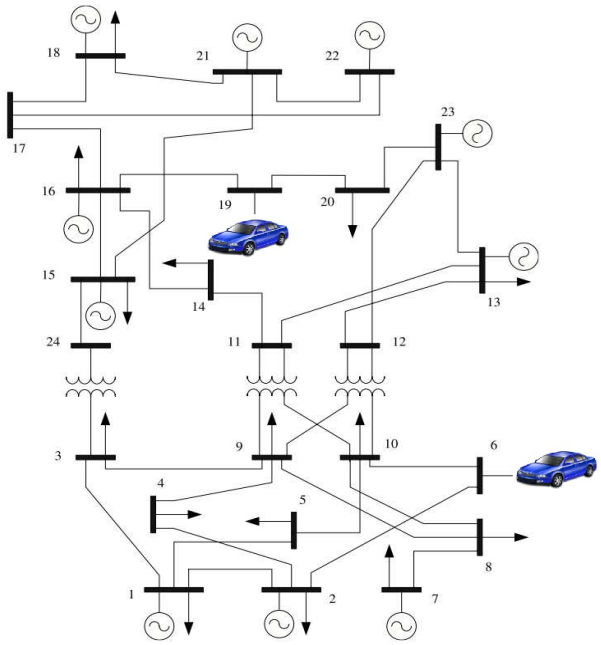


Fig.2. Modified IEEE One-Area RTS-96 system with EVs stations

According to IEC's limitation on charging current and voltage in the design standard of EVs charging system [19], we assume that both $\overline{P}_{ev.ch(ei)}$ and $\overline{P}_{ev.dis(ei)}$ are 7.2kW. After the introduction of EVs charging/swapping stations, the restoration plan of the power system after the extreme natural disaster will be related to the time of the disaster. Multiple scenarios can be simulated for resilience enhancement in the power system. Based on the uncertain maximum number of damaged transmission lines, the following five different worst weather scenarios are simulated. The related parameters of different scenarios are set as shown in Table I to Table V. The proposed solution algorithm is implemented in MATLAB with GUROBI on a computer with an Intel Core i7 2.60 GHz CPU and 8GB of memory.

Table I
Related Coefficient for k=1

k	$N_{EV.avail}^t$	$E_{EVS.avail}^t / E_{EVS.total}$	η_{ch}^t	η_{dis}^t
1	5000	65%	60%	80%

Table II
Related Coefficient for k=3

k	$N_{EV.avail}^t$	$E_{EVS.avail}^t / E_{EVS.total}$	η_{ch}^t	η_{dis}^t
3	4000	50%	70%	70%

Table III
Related Coefficient for k=5

k	$N_{EV.avail}^t$	$E_{EVS.avail}^t / E_{EVS.total}$	η_{ch}^t	η_{dis}^t
5	3000	45%	74%	67%

Table IV
Related Coefficient for k=7

k	$N_{EV.avail}^t$	$E_{EVS.avail}^t / E_{EVS.total}$	η_{ch}^t	η_{dis}^t
7	2500	35%	80%	59%

Table V

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Related Coefficient for k=9

k	$N^t_{EV.avail}$	$E^t_{EVS.avail} / E_{EVS.total}$	η^t_{ch}	η^t_{dis}
9	1500	20%	89%	45%

When k varies from 1 to 9, Table VI gives the corresponding load shed of the power system without EVs charging/swapping stations and with EVs charging/swapping stations. From Table I, we can see that the load shed increased dramatically as the uncertain maximum number of damaged transmission lines raised. It is obvious from Table VI that the amount of load shed decreased significantly for power system with EVs stations.

We also investigate the impact of the available number of EVs during the extreme events. From Table VI, it can be seen that if the available number of EVs is less, a larger load will be shed because of the power balance requirement.

Table VI indicates that the integrated version of the power system and EVs system is helpful to deal with extreme weather events in terms of enhancing power system resilience, and the power system resilience could be further enhanced by increasing the available number of EVs under the worst scenario.

TABLE VI
Load shed without and with EVs in the IEEE RTS-96 System under different K's

k	Load shed (MW) Without EVs	Load shed (MW) With EVs
1	64	51
3	590	496
5	1,016	790
7	1,303	1210
9	1,366	1253

VII. CONCLUSION

Introducing EVs charging/swapping stations in the power system is very helpful for the system operator to enhance the system resilience. However, each electric vehicle can be characterized by its randomness, which poses significant challenges for modelling the EVs charging/discharging. In this paper, the charging and discharging of EVs are formulated and included in the resilience enhancement model. The resilience model can handle uncertain extreme weather scenarios. Two issues are addressed in this work: (i) a comprehensive integrated model is introduced, which forms a two-stage/tri-level resilience formulation; (ii) EVs concentrated charging/swapping approaches are incorporated into the proposed model. Simulation results show that introducing EVs is a suitable choice for power system resilience enhancement. Moreover, decreased load shedding by adding EVs is also demonstrated with the case studies.

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